

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	02 October 2026
Team ID	SWTID-2026-1268
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication & Roles	Login with email and password Login through company Single Sign-On (SSO) Role assignment to users by administrator
FR-2	ACL Rule Configuration	Create an ACL on a table and select the controlling field Define the field-value condition (e.g. department = HR) Enable or disable an ACL without deleting it
FR-3	Script-Controlled Access	Write an ACL script that returns true/false using the record field value and current user Test the script on sample records before activation Reuse common logic through shared script includes
FR-4	Record Access Enforcement	Show only permitted records in list views and reports Block direct access to restricted records via URL or API Hide or mask restricted fields Display a clear “access denied” message
FR-5	Audit & Reporting	Log every allowed and denied access attempt (user, record, time, result) Export the access log as a report

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Administrators can configure an ACL and its field condition through a simple form; users receive clear, understandable access-denied messages.
NFR-2	Security	Access is denied by default unless an ACL script explicitly allows it; rules are enforced on the server for UI, list, report and API access; only authorised roles can edit ACL scripts; all decisions are logged.
NFR-3	Reliability	ACL scripts must return consistent results for the same user and record; script errors fail safe by denying access, and rule changes are version-tracked so they can be rolled back.
NFR-4	Performance	ACL evaluation adds minimal delay to page load and list queries (target under 200 ms per request); scripts avoid heavy queries inside record-level checks.
NFR-5	Availability	Access control is available whenever the platform is up (target 99.9% uptime), with no single point of failure in rule evaluation.
NFR-6	Scalability	The solution supports a growing number of users, tables and ACL rules, and large record volumes, without rewriting scripts.